

Stanley College of Engineering and Technology for Women(A)

B.E I Semester (Backlog) Examinations May-June-2023

Mathematics-I
(Common to all branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|---|-----|-----|
| 1 | Write the statement of Cauchy's nth root test | 2 M | L 1 |
| 2 | Verify Rolle's theorem for $f(x) = x^3$ in $[1,3]$ | 2 M | L 1 |
| 3 | Find first and second order partial derivatives of $x^3 + y^3 - 3axy$ | 2 M | L 1 |
| 4 | Find the Curl f if $f = xyi + yzj + zxk$ | 2 M | L 1 |
| 5 | Find the first two iterations for $x^3 - 2x - 5 = 0$ by the method of Bisection correct to three decimal places | 2 M | L 2 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|---|--|------|----|
| 6 | a. Test the convergence of the series $\sqrt[n]{n+1} - \sqrt[n]{n-1}$ | 5 M | L3 |
| | b. Test the convergence of the series $\sum (nx/(n+1))^n$ | 5 M | L3 |
| | OR | | |
| | c. Test the convergence of the series $\sum (1.2.3.....n)/(3.5.7.....(2n+1))$ | 5 M | L3 |
| | d. Examine the convergence of the series $1 - 1/\sqrt{2} + 1/\sqrt{3} - 1/\sqrt{4} + \dots$ | 5 M | L3 |
| 7 | a. Prove that $\pi/6 + 1/(5\sqrt{3}) < (\sin)^{-1}(3/5) < \pi/6 + 1/(8)$ | 5 M | L4 |
| | b. Verify Cauchy's Mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ on $(0, \pi/4)$ | 5 M | L3 |
| | OR | | |
| | c. Show that the radius of curvature at any point of the cycloid $x = (a + \sin\theta)$ and $y = a(1 - \cos\theta)$ is $4a \cos(\theta/2)$ | 5 M | L4 |
| | d. Find the evloep of a system of concentric and coaxial ellipses of constant area. | 5 M | L4 |
| 8 | a. Find the maximum and minimum distances of the point $(3,4,12)$ from the sphere $x^2 + y^2 + z^2 = 4$ | 10 M | L5 |

OR

1

P.T.O

- c. If $u = F(x-y, y-z, z-x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ 5M L3
- d. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that $(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z})^2 u = (-9)/(x+y+z)^2$ 5M L4
- 9 a. If $F = (x+y+1)i + j - (x+y)k$, show that $\text{Curl } F = 0$ 5M L3
- b. Show that the vector $F = 3y^4z^2i + 4x^3z^2j + 3x^2y^2k$ is solenoidal. 5M L3
- OR
- c. Verify Green's theorem for $\int_C (xy + y^2)dx + x^2dy$ where C is bounded by $y = x$ and $y = x^2$ 10M L4
- 10 a. Apply Gauss elimination method to solve the equations $x + 4y - z = -5$, $x + y - 6z = -12$, $3x - y - z = 4$ 5M L3
- b. Find the root of the equation $\cos x = x e^x$ using the regula-falsi method correct to three decimal places. 5M L3
- OR
- c. Find the positive root of $x^4 - x = 10$ correct to three decimal places, using Newton-Raphson method. 5M L3
- d. Find the interpolating polynomial for $(0,2)$, $(1,3)$, $(2,12)$ and $(5,147)$ using Lagrange's interpolating formula. 5M L2



Hall Ticket No-

Code: S123512

Stanley College of Engineering and Technology for Women(A)

B.E(EEE &ECE) I Semester (Backlog) Examinations May-June-2023

Engineering Chemistry

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|-----|------|-----|
| 1 List out specification of potable of drinking water. | 2 M | BTL1 | CO1 |
| 2 Define entropy and enthalpy. | 2 M | BTL2 | CO2 |
| 3 Define Conducting polymers and write its classification. | 2 M | BTL1 | CO3 |
| 4 Describe the significance of Octane number and Cetane number of a fuel. | 2 M | BTL4 | CO4 |
| 5 Explain briefly Carbon nano tubes. | 2 M | BTL2 | CO5 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|--|------|------|-----|
| 6 a. Explain the purification of water by ion exchange process and give its advantages. | 10 M | BTL2 | CO1 |
| OR | | | |
| c. What are boiler troubles? Explain formation & prevention of scale and sludge. | 5 M | BTL2 | CO1 |
| d. Demonstrate, rusting of iron "with the help of electrochemical theory of corrosion" | 5 M | BTL3 | CO1 |
| 7 a. Explain the physical significance of entropy. | 5 M | BTL2 | CO2 |
| b. Calculate the electrode potential at a copper electrode dipped in a 0.1M solution of copper sulphate at 25°C. The standard electrode potential of Cu ²⁺ /Cu system is 0.34V at 298K. | 5 M | BTL3 | CO2 |
| OR | | | |
| c. Describe the construction and working of calomel electrode with a neat diagram. | 7 M | BTL5 | CO2 |
| d. The efficiency of an engine is 0.42. Calculate the heat that must be withdrawn from the reservoir at higher temperature to produce 203 cal of work. | 3 M | BTL3 | CO2 |
| 8 a. Draw the molecular orbital diagram of N ₂ and find out the bond order. | 10 M | BTL4 | CO3 |
| OR | | | |
| c. Discuss the preparation, properties and applications of PET and Kevlar | 5 M | BTL6 | CO3 |
| d. Make use of MOT Calculate the bond order of N ₂ and CO molecule. | 5 M | BTL3 | CO3 |
| 9 a. Explain proximate analysis of coal? How is it carried out? What is its significance? | 6 M | BTL1 | CO4 |
| b. A sample of coal has the following composition: C=83%, H=7.5%, S=1.5%, N=0.6%, O=8.4% find the gross calorific value using Dulong Formula. | 4 M | BTL3 | CO4 |
| OR | | | |
| c. Explain fractional distillation of petroleum with a neat diagram. | 10 M | BTL2 | CO4 |
| 10 a. What are Lithium ion batteries? Explain the construction, working and applications of Lithium ion batteries. | 10 M | BTL5 | CO5 |
| OR | | | |
| c. Illustrate the principles of green chemistry | 10 M | BTL4 | CO5 |
